$$H|\psi_0\rangle = E_0|\psi_0\rangle$$

$$H = \underbrace{-\sum_{i=1}^{A} \frac{\nabla_i^2}{2m}}_{T} + \underbrace{\sum_{i<j} \hat{v}_{ij}}_{V} + \dots$$

$$\hat{v}_{ij} \sim \frac{e^2}{r_{ij}} \mathbb{1}$$

$$\hat{v}_{ij} = \underbrace{v^c(r_{ij})}\, \mathbb{1} + v^\sigma(r_{ij})\, \vec{\sigma}_i \cdot \vec{\sigma}_j$$

$$\vec{\sigma}_i \cdot \vec{\sigma}_j = \sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y + \sigma_i^z \sigma_j^z$$

Suppose our state is

$$\vec{\sigma}_i \vec{\sigma}_j |\uparrow\uparrow\rangle = \left(2P_{ij}^\sigma - 1\right)|\uparrow\uparrow\rangle = (2-1)|\uparrow\uparrow\rangle = |\uparrow\uparrow\rangle$$

$$\vec{\sigma}_i \vec{\sigma}_j |\uparrow\downarrow\rangle = \left(2P_{ij}^\sigma - 1\right)|\underset{\smile}{\uparrow\downarrow}\rangle = 2|\downarrow\uparrow\rangle - |\uparrow\downarrow\rangle$$

$$\vec{\sigma}_i \vec{\sigma}_j |\downarrow\uparrow\rangle = 2|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle$$

$$\vec{\sigma}_i \vec{\sigma}_j |\downarrow\downarrow\rangle = |\downarrow\downarrow\rangle$$

$$\boxed{P_{ij}^\sigma = \frac{1 + \vec{\sigma}_i \vec{\sigma}_j}{2}} \Rightarrow 2P_{ij}^\sigma - 1 = \vec{\sigma}_i \vec{\sigma}_j$$

$\langle \psi_T | H | \psi_T \rangle =$

$|\psi_T\rangle = \underbrace{|\psi_T^{\uparrow}\rangle}_{} \otimes |\psi_T^{\downarrow}\rangle$

Slater determinants

$\langle \psi_T^{\uparrow} | \otimes \langle \psi_T^{\downarrow} | \; H \; |\psi_T^{\uparrow}\rangle \, |\psi_T^{\downarrow}\rangle$

spin-independent

$\langle \psi_T^{\downarrow} | \psi_T^{\uparrow} \rangle = 0$

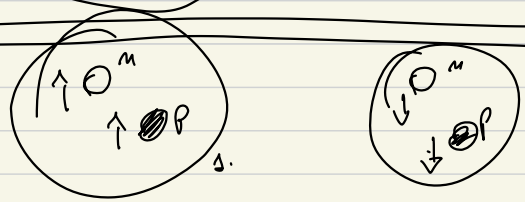
If H is spin dependent

$\left[\langle \uparrow\uparrow |_{12} \otimes \langle \downarrow\downarrow |_{34} \right] \; \underbrace{\vec{\sigma}_1 \cdot \vec{\sigma}_3}_{} \; \left[|\uparrow\uparrow\rangle_{12} \otimes |\downarrow\downarrow\rangle_{34} \right]$

$\left(2 P_{13} - 1 \right)$

$|\downarrow\uparrow\rangle \, |\uparrow\downarrow\rangle$

$^2H + n \rightarrow P, n$



$J = 1 \quad T = 0$

$J = 1 \Rightarrow S = 1$

$|\uparrow\uparrow\rangle ; \; |\downarrow\downarrow\rangle ; \; \underbrace{\frac{1}{\sqrt{2}} \left(|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle \right)}_{\text{Triplet}}$

$$H = v^c(r_{12}) + v^\sigma(r_{12})\,\vec{\sigma}_1\cdot\vec{\sigma}_2 = v^c(r_{12}) + v^\sigma(r_{12})\left[2P^\sigma_{12} - 1\right]$$

$$H\,|\uparrow\uparrow\rangle = v^c(r_{12})\,|\uparrow\uparrow\rangle + (2-1)\,v^\sigma(r_{12})\,|\uparrow\uparrow\rangle$$

$$= \left[v^c(r_{12}) + v^\sigma(r_{12})\right]|\uparrow\uparrow\rangle$$

$$H\,|\downarrow\downarrow\rangle = \left[v^c(r_{12}) + v^\sigma(r_{12})\right]|\downarrow\downarrow\rangle$$

$$H\left(\frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}\right) = \left[v^c(r_{12}) + v^\sigma(r_{12})\right]\left(\frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}\right)$$

$S=1$ what about the Isospin? $T=0$

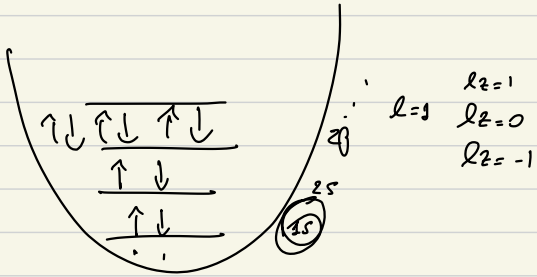
$p, n \rightarrow$ $p \leftrightarrow \uparrow$ $n \leftrightarrow \downarrow$ $n \sim n$

$T=0 \leftrightarrow \dfrac{|pn\rangle - |np\rangle}{\sqrt{2}}$ $S=1$ $S=0$

$T=1$ $\left(|pp\rangle ; |nn\rangle ; \dfrac{|pn\rangle + |np\rangle}{\sqrt{2}}\right)$

Light nuclei :

$A = 2 \quad \leftarrow \; {}^2H$



$l = 1 \quad l_z = 1, \; l_z = 0, \; l_z = -1$

2 particles ; $l = 0 \rightarrow 1s$

$\underbrace{\Phi_1}_{\text{radial}}(\vec{r}_1) \cdot |\chi_{\sigma_z}\rangle \otimes |\chi_{\tau_z}\rangle$

$R_{nl}(r_1) Y_{l l_z}(\hat{r}_1)$

$|\uparrow\rangle \quad |p\rangle$

$|\downarrow\rangle \quad |p\rangle$

$|\uparrow\rangle \quad |n\rangle$

$|\downarrow\rangle \quad |n\rangle$

$l = 0 \; l_z = 0$

$Y_{00} = 1$

$R_{00} = e^{-\alpha r^2} \; ; \; e^{-\alpha r} \; ; \ldots \; \sum c_i e^{-\alpha_i r^2} \ldots$

$$|{}^2H\rangle = \det \begin{pmatrix} \Phi_1(\vec{r}_1)|\uparrow p\rangle & \Phi_1(\vec{r}_2)|\uparrow p\rangle \\ \Phi_1(\vec{r}_1)|\uparrow n\rangle & \Phi_1(r_2)|\uparrow n\rangle \end{pmatrix}$$

states $\rightarrow$

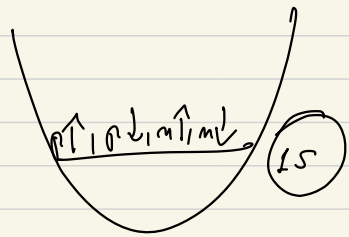
particles

$$|{}^2H\rangle = \Phi_1(\vec{r}_1)\,\Phi_1(\vec{r}_2)\left[|\uparrow_p\rangle|\uparrow_n\rangle - |\uparrow_n\rangle|\uparrow_p\rangle\right]$$

$S=1$

$T=0$

$|{}^4He\rangle$



$$|{}^4He\rangle = \det\begin{pmatrix} \Phi_{1S}(r_1)|p\uparrow\rangle_1 & \Phi_{1S}(r_2)|p\uparrow\rangle_2 & \Phi(r_3)|p\uparrow\rangle_3 & \Phi(r_4)|p\uparrow\rangle_4 \\ \Phi_{1S}(r_1)|p\downarrow\rangle_1 & \Phi_{1S}(r_2)|p\downarrow\rangle_2 & \Phi_{1S}(r_3)|p\uparrow\rangle_3 & \Phi_{1S}(r_4)|p\uparrow\rangle_4 \\ \Phi_{1S}(r_1)|n\uparrow\rangle_1 & \cdots & & \cdots \\ \cdots & & & \end{pmatrix}$$

$$= \Phi_{1S}(r_1)\Phi_{1S}(r_2)\Phi_{1S}(r_3)\Phi_{1S}(r_4)\left[|p\uparrow\rangle_1|p\downarrow\rangle_2|n\uparrow\rangle_3|n\downarrow\rangle_4 - |p\downarrow\rangle_1|p\uparrow\rangle_2|n\uparrow\rangle_3|n\downarrow\rangle_4 - \cdots\right]$$

$$|\psi\rangle = \sum_{i_s i_t} \psi(i_s, i_t)\, |i_s, i_t\rangle$$

$$2^A \times \frac{A!}{(A-Z)!\,Z!}$$

$$= \begin{cases} \psi_{\uparrow\uparrow pn} \\ \psi_{\uparrow\downarrow pn} \\ \psi_{\downarrow\uparrow pn} \\ \psi_{\downarrow\downarrow pn} \\ \psi_{\uparrow\uparrow np} \\ \psi_{\uparrow\downarrow np} \\ \psi_{\downarrow\uparrow np} \\ \psi_{\downarrow\downarrow np} \end{cases}$$

$pn \qquad np$

$A = 2 \quad Z = 1$

$$\frac{2!}{1!\,1!} = 2$$

$$2^2 \times 2$$

$$|^2H\rangle = \Phi_{1s}(r_1)\Phi_{1s}(r_2)\, \frac{1}{\sqrt{2}}\left[|p\uparrow\rangle |n\uparrow\rangle - |n\uparrow\rangle |p\uparrow\rangle \right]$$

$$\psi_{\uparrow\uparrow pn} = ! = \frac{1}{\sqrt{2}} \Phi_{1s}(r_1)\Phi_{1s}(r_2)$$

$$\psi_{\uparrow\downarrow pn} = 0$$

$$\psi_{\downarrow\uparrow pn} = 0$$

$$\psi_{\downarrow\downarrow pn} = 0$$

$$\psi_{\uparrow\uparrow np} = \frac{1}{\sqrt{2}} \Phi_{1s}(r_1)\Phi_{1s}(r_2)$$

$$\vdots \quad = 0$$

$$\sigma_{12} = \vec{\sigma}_1 \cdot \vec{\sigma}_2 = \left(2P^{\sigma}_{12} - 1\right)$$

$$\sigma_{12} \begin{pmatrix} \psi_{\uparrow\uparrow} \\ \psi_{\uparrow\downarrow} \\ \psi_{\downarrow\uparrow} \\ \psi_{\downarrow\downarrow} \end{pmatrix} = \begin{pmatrix} \psi_{\uparrow\uparrow} \\ 2\psi_{\downarrow\uparrow} - \psi_{\uparrow\downarrow} \\ 2\psi_{\uparrow\downarrow} - \psi \\ \\ \end{pmatrix}$$

$$\sigma_{12}\, \psi_{\uparrow\downarrow} |\uparrow\downarrow\rangle = 2\psi_{\uparrow\downarrow} |\downarrow\uparrow\rangle - \psi_{\uparrow\downarrow} |\uparrow\downarrow\rangle$$

$$\psi'_{\uparrow\uparrow} = \psi_{\uparrow\uparrow}$$

$$\psi'_{\uparrow\downarrow} = 2\psi_{\downarrow\uparrow} - \psi_{\uparrow\downarrow}$$

$$\psi'_{\downarrow\uparrow} = 2\psi_{\uparrow\downarrow} - \psi_{\downarrow\uparrow}$$

$$\psi'_{\downarrow\downarrow} = \psi_{\downarrow\downarrow}$$

$$\psi'_{\uparrow\uparrow} = \psi_{\uparrow\downarrow},$$

$$S_{12}$$

$$\dot{S}_{12} = \left(2P^{\sigma}_{12} - 1\right)$$

$$S_{12} |\uparrow\downarrow\rangle = \alpha |\uparrow\uparrow\rangle + \beta |\downarrow\downarrow\rangle + \gamma |\uparrow\downarrow\rangle + \delta |\downarrow\uparrow\rangle$$